

3.1

$\therefore$ 是不等长码. $\therefore$ 码字可以是 1-D 位.

即. 码字数 $\leq D + D^2 + D^3 + \dots + D^N = \frac{D(1-D^N)}{1-D}$

3.3

(a) 是前缀码. 故唯一可译.

(b) $S_0 = \{0, 01, 11\}$ $S_1 = \{1\}$ $S_2 = \{1\}$ $\dots$ $S_n = \{1\}$

$\therefore S_n \neq S_0$ $\therefore$ 唯一可译.

(c) $S_0 = \{0, 01, 11\}$ $S_1 = \{1\}$ $S_2 = \{0\}$ $0 \in S_0$

$\therefore$ 非唯一可译.

(d) $S_0 = \{0, 01\}$ $S_1 = \{1\}$

(e) 是前缀码. $\therefore$ 唯一可译.

(f) $S_0 = \{110, 11, 10\}$ $S_1 = \{0\}$ $0 \notin S_0$ $\therefore$ 唯一可译.

(g) $S_0 = \{110, 100, 00, 10\}$ $S_1 = \{0\}$ $S_2 = \{0\}$ $\therefore$ 非唯一可译.

3.4

(a) $S_1 = \{1\}$ $S_2 = \{100, 110, 01011\}$ $S_3 = \{11\}$ $S_4 = \{00, 110\}$

$S_5 = \{01, 011, 110, 0\}$ $S_6 = \{0, 10, 001, 110, 001, 0110\}$

构造的模糊序列: 000110101110001 $\begin{cases} \pi_5 \pi_1 \pi_7 \pi_2 \\ \pi_2 \pi_8 \pi_4 \pi_2 \end{cases}$

(b) $S_1 = \{d, abd\}$ $S_2 = \{ba\}$ $S_3 = \{ce\}$ $S_4 = \{ac, ab\}$ $S_5 = \{c, cd\}$

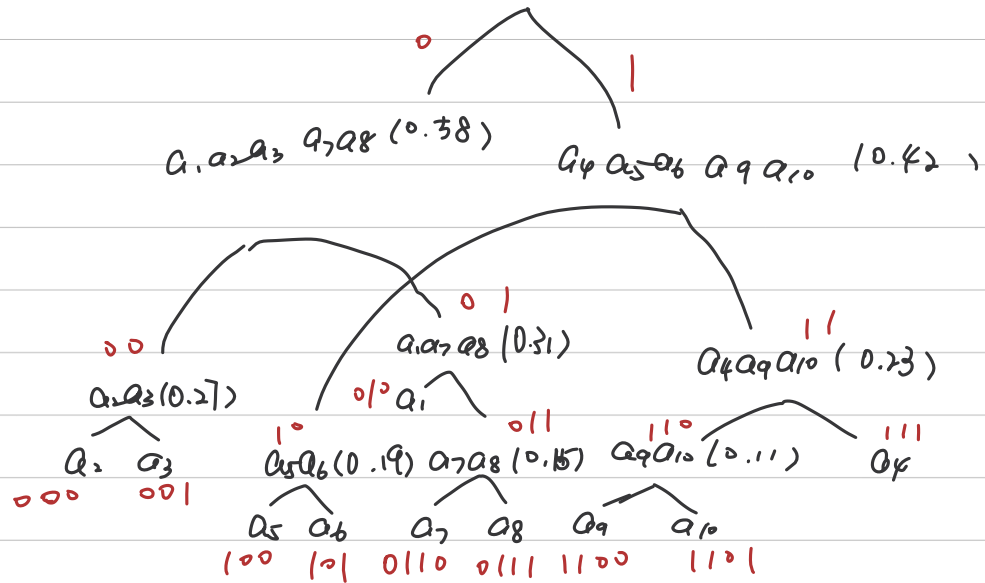
$S_6 = \{eac, eab\}$ $S_6 = \{d\}$ $S_6 = \{ba\} = S_2$

$\therefore$ 非唯一可译.

3.6

$H(u) = -\sum_{i=1}^{10} p_i \log p_i = 3.2345 \text{ bit}$

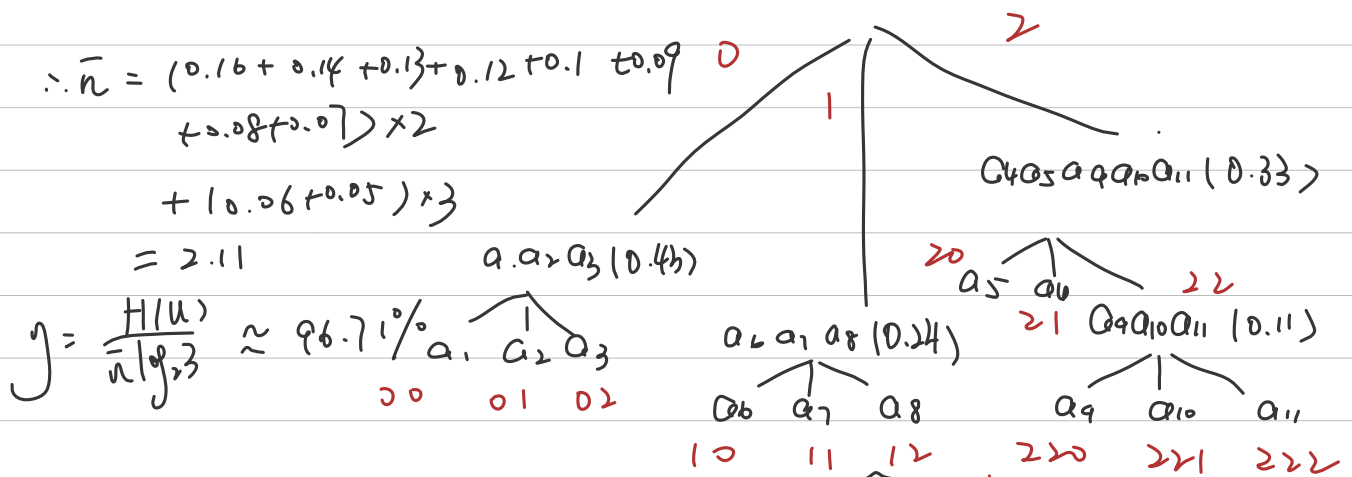
(a)



$$\therefore \bar{n} = (0.08 + 0.07 + 0.06 + 0.05) \times 4 + (0.09 + 0.1 + 0.12 + 0.13 + 0.14 + 0.16) \times 3$$

$$g = \frac{H(u)}{\bar{n} \lg 2} = 99.21\% = 3.16$$

(b) 设 $P_{a_{11}} = 0$.



$$\therefore \bar{n} = (0.16 + 0.14 + 0.13 + 0.12 + 0.1 + 0.09 + 0.08 + 0.07) \times 2 + (0.06 + 0.05) \times 3 = 2.11$$

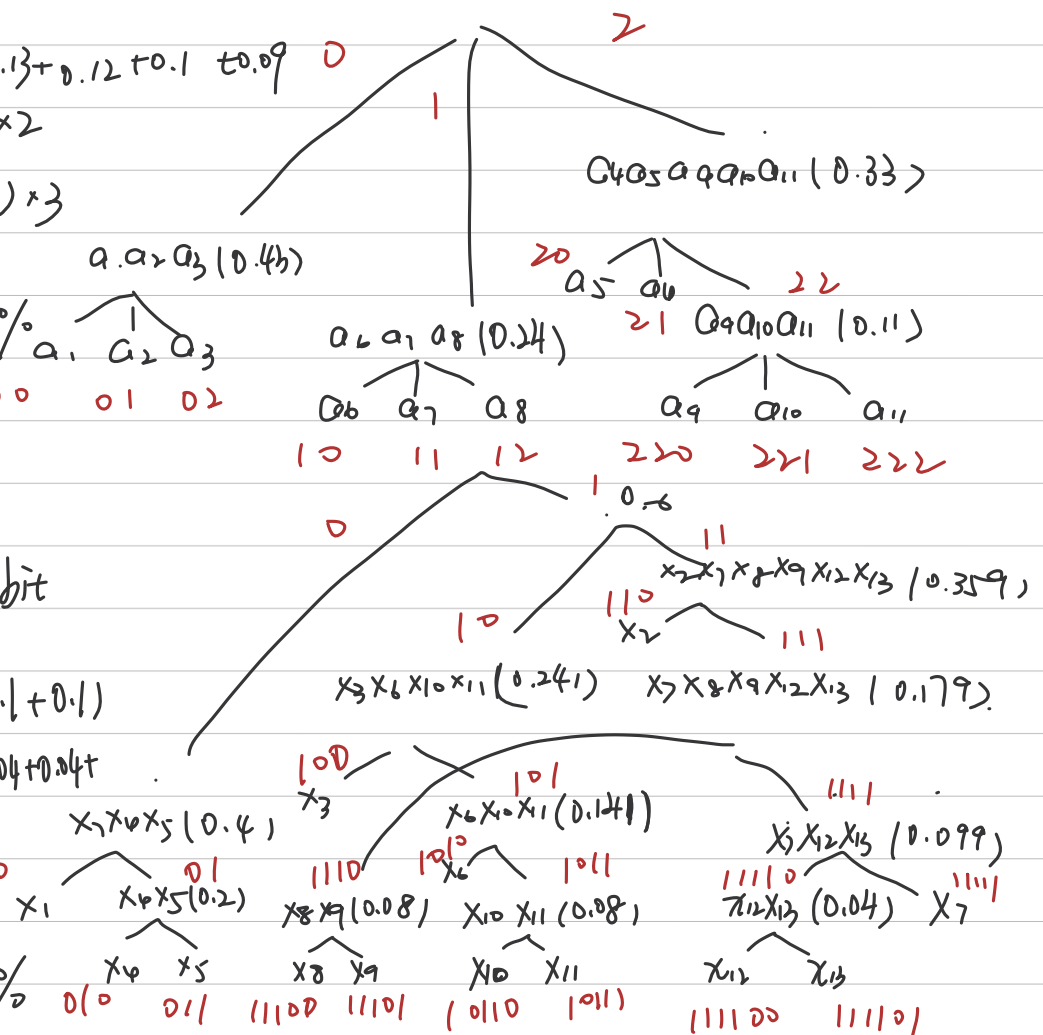
$$g = \frac{H(u)}{\bar{n} \lg 2} \approx 96.71\%$$

3.11 $H(u) = 3.3545$ bit

(a)

$$\bar{n} = 0.2 \times 2 + 3 \times (0.1 + 0.18 + 0.1 + 0.1) + 4 \times 0.06 + 5 \times (0.059 + 0.04 + 0.04 + 0.04 + 0.04) + 6 \times (0.03 + 0.01) = 3.419$$

$$g = \frac{3.3545}{3.419 \lg 2} = 98.1\%$$



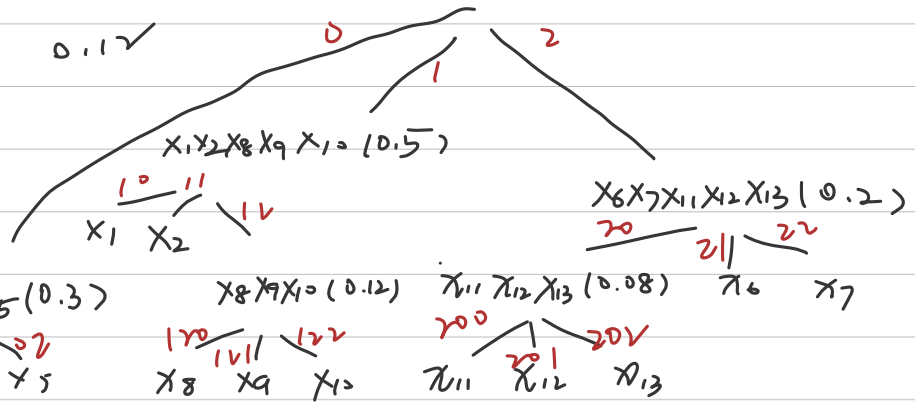
1b)

$$\therefore \bar{u} = 2 \times (0.2 + 0.18 + 0.3 + 0.12)$$

$$+ 3 \times 0.2$$

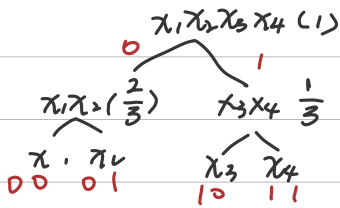
$$= 2.2$$

$$\eta = \frac{H(u)}{n \log 2} = 96.71\%$$



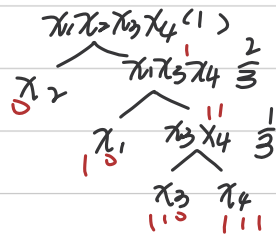
3.10.

(a)



(b) 可以有第2种编码方式

对于第1种:



$$\bar{n}_1 = 2$$

对于第2种:

$$\bar{n}_2 = 1 \times \frac{1}{3} + 2 \times \frac{1}{3} + 3 \times \frac{1}{3} = 2$$

$$\bar{n}_1 = \bar{n}_2 \quad \text{证毕}$$

(c)

对 x_3 来说, 其 Shannon 码长为 $l_3 = \lceil \log_2 \frac{1}{p_3} \rceil = 2$

而其在第2种编码中, 码长为3 > $l_3 = 2$
 $\therefore$ 可能超过 $\lceil \log_2 \frac{1}{p_{\max}} \rceil$

3.16.

状态转移矩阵 P :

$$P = \begin{bmatrix} 1 & 0 & \frac{3}{4} & \frac{1}{4} \\ 2 & 0 & \frac{1}{2} & \frac{1}{2} \\ 3 & 1 & 0 & 0 \end{bmatrix}$$

$$QP = Q$$

$$\begin{cases} p_1 + p_2 + p_3 = 1 \\ p_1 = p_3 \\ \frac{3}{4}p_1 + \frac{1}{2}p_2 = p_2 \end{cases} \Rightarrow \begin{cases} p_1 = \frac{2}{7} \\ p_2 = \frac{3}{7} \\ p_3 = \frac{2}{7} \end{cases}$$

$$p_{a_1} = p_1 \cdot p_{1,2} + p_3 \cdot p_{3,1} = \frac{2}{7} \times \frac{1}{2} + \frac{2}{7} \times 1 = \frac{3}{7}$$

$$p_{a_2} = p_1 \cdot p_{1,2} + p_2 \cdot p_{2,2} = \frac{2}{7}$$

$$p_{a_3} = 1 - \frac{2}{7} - \frac{3}{7} = \frac{2}{7}$$

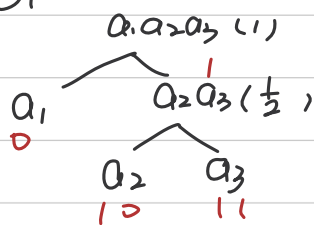
$$(b) H(U|S_1) = - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right) = 1.5 \text{ bit}$$

$$H(U|S_2) = - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) = 1 \text{ bit}$$

$$H(U|S_3) = -1 \log_2 1 = 0 \text{ bit}$$

$$(c) H_{\infty}(U) = \sum_{i=1}^3 p(i) \cdot H(U|S_i) = 0.8571 \text{ bit}$$

(d) for S_1 :



for S_2 :

$$a_2 \rightarrow 0$$

$$a_3 \rightarrow 1$$

for S_3 :

$$a_1 \rightarrow 0$$

$$(e) \bar{n}_1 = 1 \times \frac{1}{2} + 2 \times \frac{1}{4} \times 1 = 1.5$$

$$\bar{n}_2 = 1 \times \frac{1}{2} \times 2 = 1$$

$$\bar{n}_3 = 1$$

$$\therefore \bar{n} = p_1 \bar{n}_1 + p_2 \bar{n}_2 + p_3 \bar{n}_3 = \frac{8}{7}$$

